

Bluetooth Channel Sounding

Mode-2 Phase-Based Ranging

Tone exchange, the phase correction term, two-way phase cancellation, phase-slope and impulse-response range estimation, and ambiguity resolution

This report traces the complete measurement chain of a Bluetooth Channel Sounding Mode-2 step: the reciprocal exchange of unmodulated tones across $N_{AP} + 1$ tone slots, the complex phase measurement made in each slot, its reporting as a phase correction term, and the conversion of a set of per-channel phase correction terms into a distance. It derives the two-way phase construction that cancels the unknown relative local-oscillator phase, shows why that construction leaves the four hardware phase responses as an additive loop group delay, and shows why a residual fractional frequency offset from Mode-0 turns directly into a range bias through the interval between the two measurements. It develops the estimator family that serves the observable: single-tone phase and frequency estimation, weighted phase-slope regression, inverse-transform channel impulse response formation, and super-resolution first-path detection, together with the Cramér–Rao analysis that ties ranging variance to phase variance, channel count and frequency span. It closes with ambiguity resolution against Mode-1, an error budget, a layered verification plan, and the boundary between specified behaviour and implementation choice.

Primary references	Core 6.3, Vol. 6, Part H, Sec. 4.3.3; Vol. 1, Part A, Sec. 9.2
Air-interface timing	Core 6.3, Vol. 6, Part H, Secs. 4.3 and 4.5
Host interface	Core 6.3, Vol. 4, Part E, LE CS Subevent Result event
Companion reports	<i>Bluetooth CS Mode-0 Frequency Estimation</i> , Revision 4.0; <i>Bluetooth CS Mode-1 RTT Estimation</i> , Revision 1.0
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Abstract

Mode-2 is the phase-based ranging step of Bluetooth Low Energy Channel Sounding [Core 6.3, Vol. 6, Part H, Sec. 4.3.3]. The initiator transmits a sequence of unmodulated tones on the step's channel, one per antenna path, and the reflector measures the complex amplitude of each; after the interlude period T_{IP2} the roles reverse and the initiator measures the reflector's tones. Each device reports, per antenna path, a complex phase correction term. The sum of the two measured phases removes the unknown relative local-oscillator phase and leaves twice the propagation phase. Repeating this across the hopped channel set gives phase as a function of frequency, whose slope is the round-trip delay and hence the distance.

The specification defines the tone-slot structure, the permitted timings, the antenna path configuration and permutation, the reported phase correction term and tone quality indicator, and the requirement that Mode-0 precede the ranging steps [Core 6.3, Vol. 6, Part H, Secs. 4.3–4.5; Vol. 4, Part E]. It does not mandate how phase is estimated from samples, how the per-channel terms are combined, or how ambiguity is resolved: Volume 1, Part A, Section 9.2 describes distance estimation from phase and amplitude information architecturally rather than normatively. This report therefore separates normative behaviour from receiver and algorithm design.

It derives the two-way phase relation from oscillator-referenced phases, shows that the four transmit and receive chain phase responses accumulate into a single additive loop group delay that must be calibrated rather than assumed to cancel, and shows that a residual fractional frequency offset ϵ produces a range bias $c \epsilon \Delta T/2$ set by the separation ΔT between the reciprocal measurements. A dedicated section develops the phase-estimator family and the two families of range extraction — slope regression in the frequency domain and peak search in the inverse-transform domain — with their ambiguity, resolution and multipath properties. A worked example at 2440 MHz over 72 channels quantifies the achievable ranging uncertainty, the cost of an uncorrected clock scale error, and the reporting quantum. The report closes with an error budget, a layered verification plan, and the specified/implementation boundary.

Status and interpretation. Normative claims are tied to the cited Bluetooth Core Specification sections. Phase estimators, ranging algorithms, quality metrics, calibration databases and aggregation rules are representative implementation methods. Timing parameter value lists, antenna configuration indices and report field encodings should be read from the adopted specification revision in force; where this report states them, they are given to make the equations concrete, and section numbering within Volume 6, Part H has changed between Core revisions. The Bluetooth specification and adopted test specifications remain authoritative for conformance.

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1 Scope and measurement objective

Bluetooth Channel Sounding uses four step modes, defined in Core 6.3, Volume 6, Part H, Sections 4.3.1–4.3.4. Mode-0 establishes the relative frequency and timing scale between the two devices. Mode-1 uses that scale to measure a round-trip time. Mode-2 measures phase on tone exchanges. Mode-3 combines a Mode-1 packet exchange with a Mode-2 tone exchange in a single step.

The Mode-2 measurement objective is not a distance and not a phase in radians. It is, per step and per antenna path, a *complex number* — the phase correction term PCT — expressed in the measuring device’s own local-oscillator reference, together with a tone quality indicator and enough metadata to associate it with a channel and a physical antenna path [Core 6.3, Vol. 4, Part E, LE CS Subevent Result]. The engineering chain is

1. open the receive window on the step’s channel at the expected carrier derived from the Mode-0 result, and capture each tone slot;
2. estimate the complex amplitude of the tone in each slot, referenced to the local oscillator;
3. map slot index to physical antenna path through the antenna path permutation, and report PCT and the tone quality indicator;
4. at the ranging entity, form the two-way phase sum per channel and antenna path, which cancels the unknown relative oscillator phase;
5. remove the calibrated loop phase response and the residual frequency-offset term;
6. assemble the per-channel complex values into a frequency-domain channel estimate, and extract a delay by slope regression or by inverse transform with first-path detection; and
7. resolve the periodic ambiguity of the phase observable against the Mode-1 round-trip-time estimate, and aggregate.

1.1 What Mode-2 measures and what it does not

Mode-2 produces a *precise* but *ambiguous* distance estimate — the exact complement of Mode-1. Its precision follows from the synthesised aperture: measuring phase at K channels spread over a span S of up to 74 MHz gives a delay resolution of order $1/S \approx 13.5$ ns, and a thermal precision far below that, because a phase slope fitted through many high-signal-to-noise points is located to a small fraction of one fringe. Its ambiguity follows from the same periodicity: phase is known only modulo 2π , so the distance repeats every $c/(2\Delta f)$, which is 149.9 m at the 1 MHz channel spacing of the Bluetooth Low Energy channel raster.

The CS channel raster is not contiguous, and the gap matters. Channel Sounding uses 72 RF channels at $2402 + k$ MHz for $k = 2 \dots 22$ and $k = 26 \dots 76$ [Core 6.3, Vol. 6, Part A, Sec. 2]. They therefore run from 2404 to 2478 MHz — a span of 74 MHz, not 71 — and there is a 4 MHz gap between 2424 and 2428 MHz where channels 23, 24 and 25 are absent. The span sets the resolution and enters the precision bound; the largest gap sets the limit beyond which sequential phase unwrapping fails, and 4 MHz is four times the raster spacing. Section 9.1 returns to this: it reduces the safe unwrapping range from 74.95 m to 18.74 m even before any channel is excluded by the channel map.

Three properties that are frequently confused. (1) The two-way phase construction cancels the unknown *relative oscillator phase* of the two devices, but it *accumulates* the four transmit and receive chain phase responses — exactly as the Mode-1 reciprocal construction cancels the reflector turnaround but accumulates the four chain group delays. (2) A residual fractional frequency offset does not merely add noise: because the residual offset in hertz scales with the channel frequency, it produces a phase term *linear in frequency*, which is indistinguishable from range. (3) Multipath error in Mode-2 is *not* one-sided. The phase slope of a superposition is an amplitude-weighted quantity that can be biased short as easily as long, which is why first-path detection in the transform domain matters more here than a robust combiner does.

1.2 Why the observable is reported, not the range

The controller reports PCT values; it does not report a distance. Core 6.3, Volume 1, Part A, Section 9.2 describes distance estimation from phase and amplitude information as an architectural capability, and the mapping from a set of phase correction terms to a metre value is deliberately left outside the specification. This is the single most important structural fact about Mode-2 for an implementer: the entire accuracy-determining part of the chain — phase estimation, calibration, transform, first-path detection, ambiguity resolution, aggregation — is design freedom, and therefore is where accuracy is won or lost.

2 Normative framework and terminology

Table 1: Normative map for the Mode-2 phase-based ranging measurement. Section numbering should be confirmed against the specification revision in force.

Source	What it defines	Question answered
Core 6.3, Vol. 1, Part A, Sec. 9.1	CS procedure architecture; ordering of calibration and ranging steps	Why must Mode-0 precede Mode-2?
Core 6.3, Vol. 1, Part A, Sec. 9.2	Distance estimation based on phase and amplitude information	What is the intended use of the reported phase?
Core 6.3, Vol. 6, Part H, Sec. 4.3.3	Mode-2 step structure: tone slots, ordering, antenna paths	What is transmitted, in which order, and when?
Core 6.3, Vol. 6, Part H, Sec. 4.3	T_{SW} , T_{PM} , T_{RD} , T_{IP2} , T_{FCS} and permitted value sets	Which step timings may be configured?
Core 6.3, Vol. 6, Part H, Sec. 4.5	Antenna path handling, permutation, tone extension slot	Which physical path does a slot correspond to?
Core 6.3, Vol. 6, Part A, Sec. 3.5	Frequency measurement and generation in CS; expected frequencies for non-Mode-0 transmissions	On what carrier is the Mode-2 tone expected?
Core 6.3, Vol. 6, Part H, Sec. 4.2	CS channel selection and hopping within a procedure	Which channels form the aperture, and in what order?
Core 6.3, Vol. 4, Part E, LE CS Subevent Result event	Per-step PCT and tone quality reporting	How is the measurement exposed to the host?
Core 6.3, Vol. 4, Part E, HCI CS configuration commands	Antenna configuration index, T_{IP2} , T_{PM} , T_{SW} , mode selection	How is the step parameterised?
RFPHY.TS / LL.TS, CS step Mode-2 clauses	Instrument-level verification of tone timing and phase reporting	How is the behaviour tested?

Table 2: Principal symbols.

Symbol	Meaning	Units
f_k	Carrier frequency of CS channel k	Hz
Δf	Channel spacing of the CS raster (1 MHz)	Hz
S	Frequency span actually measured, $f_{max} - f_{min}$	Hz
K	Number of channels contributing to one range estimate	—
τ	One-way propagation delay, d/c	s
$\theta_R(f_k)$	Phase measured at the reflector on channel k	rad
$\theta_I(f_k)$	Phase measured at the initiator on channel k	rad
$\Theta(f_k)$	Two-way phase sum $\theta_I + \theta_R$	rad
ϕ_I, ϕ_R	Local-oscillator phases of the two devices	rad
PCT	Reported phase correction term, complex	—
$H[k]$	Assembled frequency-domain channel estimate	—
$h(t)$	Inverse-transform channel impulse response	—
τ_ℓ, a_ℓ	Delay and complex gain of multipath component ℓ	s, —
D_{loop}	Total calibrated loop group delay of the four chains	s
$\Phi_{loop}(f)$	Total calibrated loop phase response	rad
ϵ	Residual fractional frequency offset after Mode-0	—
ΔT	Interval between the reciprocal measurements of one path	s
T_{SW}	Antenna path switch period	s
T_{PM}	Phase measurement period	s
T_{RD}	Transmitter ramp-down period	s
T_{IP2}	Interlude period between initiator and reflector tone sequences	s
N_{AP}	Number of antenna paths	—
TQI	Tone quality indicator	index

3 Position of Mode-2 in the CS subevent

Figure 1 places Mode-2 in the causal chain. Mode-0 runs first and produces the subevent frequency reference FFO_{RX}^* [Core 6.3, Vol. 6, Part H, Sec. 4.3.1]. That reference is used for two distinct purposes in Mode-2: it places the expected carrier so the tone is received close to the centre of the measurement filter, and it bounds the residual frequency offset whose product with the reciprocal measurement interval is a direct range bias.

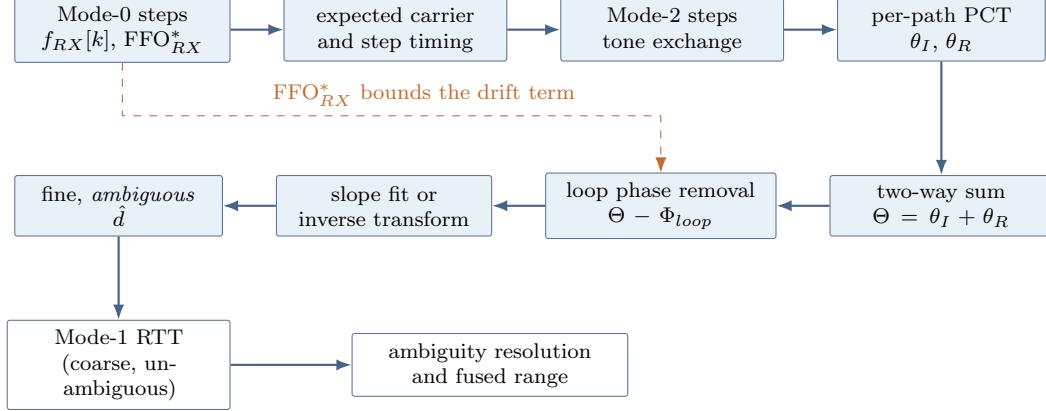


Figure 1: Position of Mode-2. Mode-0 supplies both the expected carrier and the bound on the residual frequency offset that enters the two-way phase. Mode-2 produces a precise but periodic range which is disambiguated by the Mode-1 estimate.

3.1 Why Mode-2 cannot precede Mode-0

Three independent effects make the ordering mandatory rather than conventional, and each is stronger than the corresponding Mode-1 effect.

Carrier placement. The phase measurement is made over a window of T_{PM} , typically $10\mu\text{s}$. A residual carrier offset δf rotates the tone by $2\pi\delta f T_{PM}$ within the window. An uncorrected 12 ppm crystal pair at 2.44 GHz gives $\delta f = 29.3\text{ kHz}$ and therefore 1.84 rad of rotation *inside a single measurement*, which destroys a naive coherent phase estimate outright. Mode-0 exists so that the residual is small enough for the tone to be treated as stationary in phase over T_{PM} , or at worst as a small known linear ramp.

Reciprocal interval drift. The reflector measures the initiator’s tone at one instant and the initiator measures the reflector’s tone about ΔT later. Any residual frequency difference accumulates $2\pi\delta f \Delta T$ of relative oscillator phase in between, and that term does *not* cancel in the two-way sum. Section 6.4 shows that because δf scales with the channel frequency, this term is linear in frequency and therefore reads directly as distance: $c\epsilon\Delta T/2$, which is 34 cm for an uncorrected 12 ppm pair and $\Delta T = 190\mu\text{s}$.

Frequency raster. The tone must be generated at the channel’s nominal centre as seen by the peer [Core 6.3, Vol. 6, Part A, Sec. 3.5]. A device that has not measured the peer’s frequency error places its tone at a frequency the peer does not expect, which biases the measurement by the frequency-dependent part of the receive chain response and, in a narrow measurement filter, by filter phase slope as well.

4 Mode-2 timing and the tone-slot structure

4.1 One complete Mode-2 step

Figure 2 shows the air-interface order of a Mode-2 step [Core 6.3, Vol. 6, Part H, Sec. 4.3.3]. Widths are labelled rather than drawn to scale, because the tone-sequence duration depends on the antenna configuration and on the configured T_{SW} and T_{PM} , and T_{IP2} is configured.

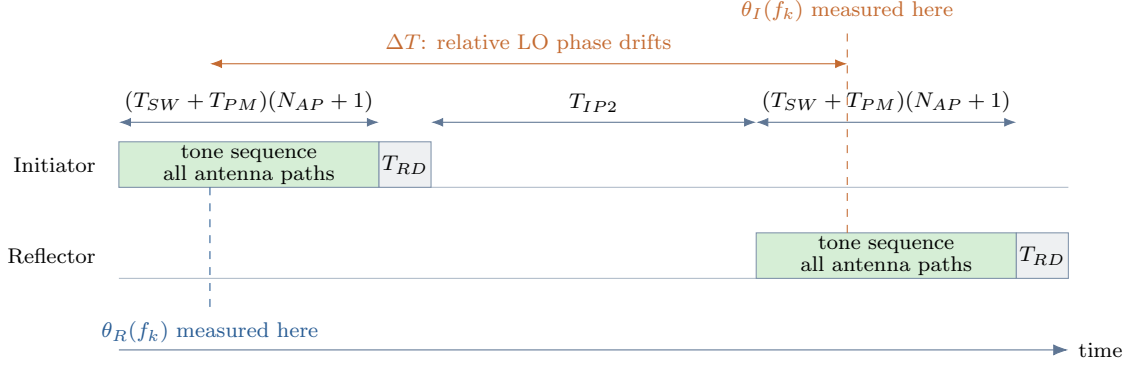


Figure 2: Mode-2 step. The initiator transmits a tone sequence covering every antenna path, ramps down, and after the interlude period T_{IP2} the reflector transmits its own tone sequence on the same channel. Each device measures the phase of the tone it receives against its own local oscillator. The interval ΔT between the two measurements of a given antenna path is the interval over which the relative oscillator phase is free to drift.

The nominal step duration is

$$T_{step,2} = 2[(T_{SW} + T_{PM})(N_{AP} + 1) + T_{RD}] + T_{IP2}, \quad (1)$$

with the frequency-change period T_{FCS} inserted between scheduled steps when consecutive steps use different channels [Core 6.3, Vol. 6, Part H, Sec. 4.3]. Note the structural difference from Mode-1: there is no packet, so there is no T_{SY} ; the transmission is a continuous unmodulated carrier, and the step length is set by the number of antenna paths rather than by a sequence length.

The interval that matters for the frequency-error term is not $T_{step,2}$ but the separation between the two measurements of the *same* antenna path,

$$\Delta T = (T_{SW} + T_{PM})(N_{AP} + 1) + T_{RD} + T_{IP2}, \quad (2)$$

which for identical slot ordering in both directions is independent of which path is considered.

4.2 The tone sequence and one tone slot

Each tone sequence consists of $N_{AP} + 1$ slots. Each slot is a switch period T_{SW} , during which the antenna switch changes state and the radio settles, followed by a phase measurement period T_{PM} during which a continuous unmodulated tone at the channel centre frequency is transmitted and the peer makes its measurement [Core 6.3, Vol. 6, Part H, Secs. 4.3.3 and 4.5].

T_{SW} is not dead time. The switch period exists so that the antenna switch, any tuning network and the receive chain reach a settled state before the measurement window opens. Two implementation consequences follow. First, the local oscillator must run *continuously* across the whole tone sequence: if it is retuned, restarted or re-locked between slots, the antenna paths no longer share a common phase reference and the per-path terms cannot be compared. Second, T_{SW} must be chosen against the measured settling time of the actual switch and filter, not against the minimum permitted value; residual settling inside T_{PM} appears as a slot-dependent phase bias that looks exactly like an antenna calibration error.

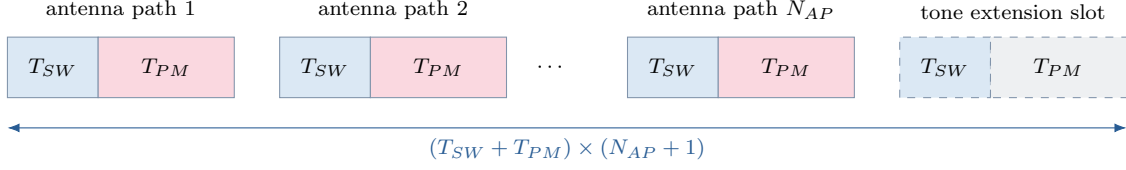


Figure 3: Tone sequence structure within one transmit slot. The first N_{AP} slots carry the tone on successive antenna paths; the final slot is the tone extension slot, whose transmission is configurable and whose reception gives a same-step estimate of the noise and interference floor.

Table 3: Mode-2 step timing parameters. Permitted sets are configuration-dependent and are subject to both devices' declared capabilities; read the value lists from the specification revision in force [Core 6.3, Vol. 6, Part H, Sec. 4.3; Vol. 4, Part E, CS configuration].

Parameter	Meaning	Representative permitted values
T_{SW}	Antenna path switch period, including RF and phase settling	1, 2, 4, 10 μs when switching is performed; 0 with a single antenna
T_{PM}	Phase measurement period	10, 20, 40 μs (40 μs mandatory)
T_{RD}	Transmitter ramp-down	5 μs
T_{IP2}	Interlude period, end of initiator ramp-down to start of reflector tone sequence	10, 20, 30, 40, 50, 60, 80, 145 μs
T_{FCS}	Frequency-change period between steps	15, 20, 30, 40, 50, 60, 80, 100, 120, 150 μs

4.3 Step durations for representative configurations

Table 4: Tone-sequence and step durations from Eqs. (1) and (2), with $T_{RD} = 5 \mu s$ and $T_{IP2} = 145 \mu s$.

N_{AP}	T_{SW}	T_{PM}	Tone sequence	$T_{step,2}$	ΔT
1	10 μs	10 μs	40 μs	235 μs	190 μs
1	2 μs	10 μs	24 μs	203 μs	174 μs
2	2 μs	10 μs	36 μs	227 μs	186 μs
4	2 μs	10 μs	60 μs	275 μs	210 μs
4	10 μs	20 μs	150 μs	455 μs	300 μs
4	2 μs	10 μs	60 μs 140 μs ($T_{IP2} = 10 \mu s$)		75 μs

Table 4 gives the tone-sequence duration, the step duration and the reciprocal measurement interval for representative parameter sets. Its last two rows carry the design message of this section: they differ only in T_{IP2} and T_{PM} , yet their ΔT values differ by a factor of four. Every microsecond added to ΔT multiplies the residual frequency error into range through Eq. (13). A configuration chosen for antenna coverage and phase averaging is simultaneously a configuration that relaxes or tightens the Mode-0 accuracy requirement by a factor of four.

5 Antenna paths, permutation, and the tone extension slot

5.1 Antenna configuration

The number of antenna paths is not chosen directly; it follows from the antenna configuration index agreed for the procedure, which states how many antenna elements each side contributes [Core 6.3, Vol. 6, Part H, Sec. 4.5; Vol. 4, Part E, CS configuration]. Each path is one (initiator element, reflector element) pair, and every path is measured in both directions within the step.

Table 5: Antenna configuration index and the resulting number of antenna paths, per [Core 6.3, Vol. 6, Part A, Sec. 5.3]. Device A is the initiator and device B the reflector; N_{AP} is bounded to $1 \leq N_{AP} \leq 4$.

ACI	Initiator	Reflector	N_{AP}	ACI	Initiator	Reflector	N_{AP}
0	1	1	1	4	1	2	2
1	2	1	2	5	1	3	3
2	3	1	3	6	1	4	4
3	4	1	4	7	2	2	4

5.2 Antenna path permutation

The order in which antenna paths occupy the tone slots is not fixed across steps. An antenna path permutation index is applied per step and is reported alongside the measurement, so that the ranging entity can map slot index back to physical path [Core 6.3, Vol. 6, Part H, Sec. 4.5; Vol. 4, Part E, LE CS Subevent Result]. Permuting the order decorrelates any slot-position-dependent bias — switch settling, thermal transients, oscillator pulling from the preceding slot — from the physical path, so that such biases average across steps instead of accumulating into a per-path offset.

A permutation error is silent and fatal. If the ranging entity applies the wrong permutation, it attributes each channel's phase to the wrong antenna path. The per-path phase-versus-frequency sequences then interleave contributions from physically different paths with different geometric delays, and the slope fit returns a plausible-looking number that is simply wrong. There is no internal consistency check that catches this in a single-path geometry, so it must be verified explicitly against a known asymmetric antenna layout.

5.3 The tone extension slot

The final slot of each sequence is the tone extension slot. Whether it is actually transmitted is configurable, and the receiving device reports whether a tone was present [Core 6.3, Vol. 6, Part H, Sec. 4.3.3; Vol. 4, Part E]. It serves two purposes that are worth separating.

- **In-step noise and interference floor.** A slot in which the peer is known not to transmit gives a measurement of everything that is not the tone, taken on the same channel, with the same gain state, microseconds away from the signal measurement. That is a far better noise reference than any out-of-step estimate, and it is the natural denominator for a per-tone signal-to-noise estimate and hence for the tone quality indicator.
- **Asymmetric capability handling.** A device that needs additional settling or that supports fewer paths can use the extension slot to keep the two sequences aligned in duration, which keeps ΔT common to both directions.

An implementation that discards the extension slot measurement is discarding the only same-step, same-channel, same-gain noise reference the step provides.

5.4 Multiple paths as independent observables

With $N_{AP} > 1$, one step yields N_{AP} two-way phase values on the same channel. They are not redundant: the paths have different geometric lengths, different antenna phase centres and different multipath realisations. Three uses are common, and they are different algorithms rather than variations of one.

1. **Diversity.** Select or weight paths by tone quality to reject a path that is in a deep fade on that channel. This is the most robust use and the cheapest.
2. **Averaging.** Combine per-path range estimates after removing the per-path antenna offsets from the calibration table. This reduces variance but is only valid once the offsets are known to better than the precision being claimed.
3. **Angle information.** The phase differences *between* paths on one channel carry the direction of arrival, exactly as in a conventional array. This is outside the ranging chain but shares the calibration.

6 From two tones to a two-way phase

6.1 Oscillator-referenced phases

Let the two devices run local oscillators whose phases at time t on channel k are

$$\varphi_I(t) = 2\pi f_k t + \phi_I, \quad \varphi_R(t) = 2\pi f_k(1 + \epsilon)t + \phi_R, \quad (3)$$

where ϵ is the residual fractional frequency offset that survives the Mode-0 correction and ϕ_I, ϕ_R are unknown constants. Set $\epsilon = 0$ for the moment; Section 6.4 restores it.

The initiator transmits its tone during the slot for antenna path p ; the reflector receives it after the propagation delay τ and measures the phase of the received tone against its own oscillator. Ignoring hardware for the moment,

$$\theta_R(f_k) = -2\pi f_k \tau + (\phi_I - \phi_R). \quad (4)$$

After ΔT the roles reverse and the initiator measures

$$\theta_I(f_k) = -2\pi f_k \tau + (\phi_R - \phi_I). \quad (5)$$

6.2 The cancellation

Adding Eqs. (4) and (5),

$$\Theta(f_k) \equiv \theta_I(f_k) + \theta_R(f_k) = -4\pi f_k \tau, \quad (6)$$

and the unknown relative oscillator phase has cancelled exactly. This is the phase-domain analogue of the Mode-1 reciprocal timestamp construction: neither device needs to know the other's oscillator phase, only what it measured against its own.

6.3 Hardware phase responses accumulate

Devices do not measure phase at the antenna. Each measurement includes the transmit-chain phase response of the transmitting device and the receive-chain phase response of the measuring device. Writing $\Phi_{TX,X}(f)$ and $\Phi_{RX,X}(f)$ for device X ,

$$\theta_R(f_k) = -2\pi f_k \tau + (\phi_I - \phi_R) - \Phi_{TX,I}(f_k) - \Phi_{RX,R}(f_k), \quad (7)$$

$$\theta_I(f_k) = -2\pi f_k \tau + (\phi_R - \phi_I) - \Phi_{TX,R}(f_k) - \Phi_{RX,I}(f_k), \quad (8)$$

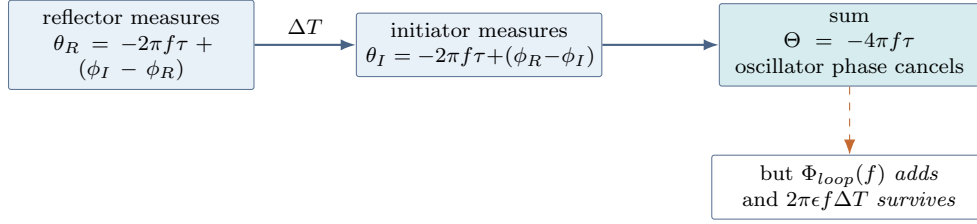


Figure 4: The two-way phase construction. The unknown relative oscillator phase cancels in the sum; the hardware phase responses and the residual-frequency drift term do not.

so that

$$\Theta(f_k) = -4\pi f_k \tau - \underbrace{[\Phi_{TX,I} + \Phi_{RX,R} + \Phi_{TX,R} + \Phi_{RX,I}]}_{\Phi_{loop}(f_k)}(f_k). \quad (9)$$

To the extent that Φ_{loop} is linear in frequency it is a pure delay, $\Phi_{loop}(f) \approx \Phi_0 + 2\pi f D_{loop}$, and it is then indistinguishable from range:

$$\hat{d}_{biased} = d + \frac{c D_{loop}}{2}. \quad (10)$$

The same result as Mode-1, reached by a different route. Equation (10) is numerically identical to the Mode-1 loop-delay bias: an uncalibrated D_{loop} of 10 ns produces 1.50 m of range error, independent of true distance and of signal-to-noise ratio, and no amount of averaging removes it. The difference is that Mode-2 is sensitive to the *whole shape* of $\Phi_{loop}(f)$, not only to its slope: any curvature — from filter roll-off at the band edges, from a VCO tuning-dependent phase, from an antenna’s own phase response — appears as a frequency-dependent residual that corrupts the slope fit and smears the inverse transform. A Mode-2 calibration is therefore a *phase response over frequency*, per antenna path and per gain state, not the single scalar that suffices for Mode-1.

The practical form of the correction is

$$\Theta_{cal}(f_k) = \text{wrap}(\Theta(f_k) + \hat{\Phi}_{loop}(f_k)), \quad (11)$$

applied before any unwrapping, with $\hat{\Phi}_{loop}$ read from a calibration surface indexed by channel, antenna path, gain state and temperature. The constant term Φ_0 affects the intercept rather than the slope, and is therefore harmless to a slope estimator but not to an inverse-transform estimator, which sees it as a complex rotation of the impulse response.

6.4 Residual frequency offset: the term that reads as distance

Restoring ϵ in Eq. (3), the relative oscillator phase is no longer constant between the two measurements. Over the interval ΔT of Eq. (2) it advances by $2\pi\epsilon f_k \Delta T$, and this term appears in the sum with a single sign:

$$\Theta(f_k) = -4\pi f_k \tau - \Phi_{loop}(f_k) + 2\pi\epsilon f_k \Delta T. \quad (12)$$

Because ϵ is a *fractional* offset, the term is proportional to f_k . It is therefore linear in frequency, exactly like the propagation term, and a slope estimator cannot separate the two. Differentiating and converting to distance,

$$\delta d_\epsilon = \frac{c \epsilon \Delta T}{2}. \quad (13)$$

Three design conclusions follow, and they differ sharply from the Mode-1 case. First, the accuracy Mode-0 must deliver for Mode-2 is roughly an order of magnitude more demanding

Table 6: Range bias from residual fractional frequency offset, Eq. (13).

Residual offset	$\Delta T = 75 \mu s$	$\Delta T = 190 \mu s$	$\Delta T = 300 \mu s$
12 ppm (uncorrected, typical crystal pair)	13.5 cm	34.2 cm	54.0 cm
1 ppm (crude correction)	1.12 cm	2.85 cm	4.50 cm
0.1 ppm	1.12 mm	2.85 mm	4.50 mm
0.05 ppm (good Mode-0 result)	0.56 mm	1.42 mm	2.25 mm

than for Mode-1, because Mode-2's own thermal floor is an order of magnitude lower: 1 ppm is adequate for Mode-1 and is already the dominant error for Mode-2. Second, a short T_{IP2} and a compact tone sequence are directly worth range accuracy. Third, the term is a *bias common to every channel in the subevent*, so averaging across channels does not reduce it — only the Mode-0 correction does.

6.5 Distance from the two-way phase

Neglecting the corrected terms, Eq. (6) gives the fundamental relation

$$\Theta(f) = -\frac{4\pi f d}{c}, \quad d = -\frac{c}{4\pi} \frac{\partial \Theta}{\partial f}, \quad (14)$$

and the two constants worth memorising are the phase step per channel and the ambiguity period:

$$\frac{\partial \Theta}{\partial f} \Delta f = \frac{4\pi \Delta f}{c} d = 0.041917 \text{ rad/m at } \Delta f = 1 \text{ MHz}, \quad d_{amb} = \frac{c}{2\Delta f} = 149.90 \text{ m}. \quad (15)$$

Equivalently, one metre of distance advances the two-way phase by 2.402° per megahertz of channel separation, and the observable repeats every 149.90 m. Unwrapping between *adjacent* channels is safe only while the step is below π , that is while

$$d < \frac{c}{4\Delta f_{max}}, \quad (16)$$

where Δf_{max} is the *largest* spacing between consecutive measured channels, not the raster pitch. For adjacent 1 MHz channels this is 74.95 m, but the CS raster's own 4 MHz gap reduces it to 18.74 m, and any further exclusion by the channel map reduces it again.

7 Signal-level model: tone to complex amplitude

7.1 Received tone

During T_{PM} the peer transmits an unmodulated carrier. Over a multipath channel with L components,

$$r(t) = \Re \left\{ \sum_{\ell=1}^L a_\ell e^{j(2\pi f_k(t-\tau_\ell)+\phi_{tx})} \right\} + n(t), \quad (17)$$

with $\tau_1 \leq \dots \leq \tau_L$ so that τ_1 is the first-arriving path. After quadrature downconversion against the local oscillator and sampling,

$$z[n] = A e^{j(2\pi \delta f n T_s + \theta)} + w[n], \quad n = 0, \dots, N-1, \quad N = T_{PM}/T_s, \quad (18)$$

where the complex amplitude $Ae^{j\theta}$ is the *sum over paths*

$$Ae^{j\theta} = \sum_{\ell=1}^L a_{\ell} e^{-j2\pi f_k \tau_{\ell}} = H(f_k), \quad (19)$$

δf is the residual carrier offset left after Mode-0, and $w[n]$ is complex noise of variance σ_w^2 per sample.

Equation (19) is the central identity of Mode-2 and deserves to be stated plainly: the quantity measured in one tone slot on one channel is one frequency-domain sample of the channel transfer function. Everything downstream is the reconstruction of a channel response from a set of such samples.

7.2 The estimation problem

Problem statement. Given samples $z[n]$ from one tone slot, estimate the complex amplitude $Ae^{j\theta}$ of the tone in the presence of an unknown small residual frequency offset δf , receiver DC offset, and I/Q imbalance; report it as a phase correction term together with a quality indicator. Then, given the set $\{\Theta(f_k)\}$ over channels, estimate the delay τ_1 of the first-arriving component and convert it to a distance.

The problem splits cleanly into a per-slot estimation problem and a cross-channel estimation problem, and the two have entirely different characters. The per-slot problem is a classical single-tone parameter estimation with an excellent signal-to-noise ratio and a very short record. The cross-channel problem is a spectral-estimation problem on non-uniformly sampled, wrapped, hopped data. Most implementations lose more accuracy in the second than in the first.

8 Per-slot phase estimators

None of this section is mandated by the specification; all of it is constrained by it, in that the tone is unmodulated, the observation length is T_{PM} , the carrier is fixed by the channel and the reporting format is fixed by the host interface.

8.1 Coherent average and the DFT bin

The maximum-likelihood estimate of $Ae^{j\theta}$ for known δf is the coherent average after dero-
tation:

$$\widehat{Ae^{j\theta}} = \frac{1}{N} \sum_{n=0}^{N-1} z[n] e^{-j2\pi\delta f n T_s}, \quad \hat{\theta} = \arg(\widehat{Ae^{j\theta}}). \quad (20)$$

With δf unknown, evaluating Eq. (20) on a grid of candidate offsets is exactly a discrete Fourier transform of the slot, and the estimate is taken at the maximum-magnitude bin. Zero padding refines the frequency grid; the phase must be read at the refined maximum, not at the nearest coarse bin, because a half-bin frequency error rotates the phase by up to π across the record.

Coherent loss. An uncorrected offset δf attenuates the coherent sum by $|\text{sinc}(\delta f T_{PM})|$ and, more importantly, biases the phase by $\pi \delta f T_{PM}$ — half the total rotation, since the average sits at the record centre. At $T_{PM} = 10 \mu s$, a 1 kHz residual gives 0.0157 rad of phase bias. This is the mechanism that converts a Mode-0 error into a Mode-2 error *within* a slot,

distinct from the ΔT mechanism of Section 6.4, and it is largely common-mode between the two directions and therefore partially cancels in the sum.

8.2 Frequency estimation before phase

Because the phase estimate is only as good as the derotation, a well-designed slot estimator estimates δf first. Three standard estimators apply directly.

Kay’s estimator. A weighted average of successive phase increments,

$$\widehat{\delta f} = \frac{1}{2\pi T_s} \sum_{n=1}^{N-1} u_n \arg(z[n]z^*[n-1]), \quad u_n \propto n(N-n), \quad (21)$$

which attains the Cramér–Rao bound at high signal-to-noise ratio and costs one pass.

Fitz / Luise–Reggiannini. Average the autocorrelation over several lags and take the phase of the weighted sum; more robust than Eq. (21) at moderate signal-to-noise ratio, at the cost of a bounded acquisition range set by the maximum lag.

Rife–Boorstyn. Coarse DFT peak followed by a local Newton refinement of the periodogram; this is the maximum-likelihood estimator for a single tone in white noise and is the natural reference implementation.

Do not estimate δf independently in every slot. The residual offset is a property of the oscillator pair over the subevent, not of the slot. Estimating it per slot adds an independent noisy quantity to each phase estimate — and, since the derotation bias is $\pi \widehat{\delta f} T_{PM}$, injects $\pi T_{PM} \sigma_{\delta f}$ of extra phase noise per slot. Estimate it jointly across the slots of a step, or carry it from Mode-0, and use the per-slot estimate only as a consistency check.

8.3 Front-end impairments

Three impairments bias a tone phase measurement in ways that a modulated-packet receiver largely escapes, because a tone sits at a single frequency and does not average them out.

DC offset. A receiver DC offset adds a constant to $z[n]$ and therefore a vector to the coherent average. If the tone is placed at exactly zero intermediate frequency, the offset is indistinguishable from the tone. Two standard mitigations are to place the tone at a deliberate non-zero intermediate frequency so that the DC term falls in a different DFT bin, and to measure the offset in the tone extension slot and subtract it. The first is strongly preferable.

I/Q imbalance. Gain and phase mismatch between the in-phase and quadrature paths creates an image at $-\delta f$. Its interference with the wanted tone produces a phase error that depends on δf and on the image rejection ratio; at 30 dB of image rejection the worst-case phase error is about $\arcsin(10^{-1.5}) = 1.8^\circ$, which is 0.031 rad — comparable to the entire thermal phase budget of Section 10. Image rejection is therefore a ranging specification, not only a sensitivity specification, and residual imbalance should be calibrated per gain state.

Phase noise. Oscillator phase noise integrated over T_{PM} adds directly to θ . Unlike thermal noise it is correlated between the slots of one sequence, so it does not average down across antenna paths within a step, though it does average across steps on different channels.

8.4 Quality metrics and the tone quality indicator

The per-slot estimator should emit, alongside PCT, at least: the estimated tone amplitude, the residual power after subtracting the fitted tone, the extension-slot noise reference, the estimated δf and its deviation from the subevent value, and any gain-change or clipping flag. The tone quality indicator reported to the host is a small ordinal summary of these [Core 6.3, Vol. 4, Part E, LE CS Subevent Result]; the mapping is an implementation characterisation.

Table 7: Reported tone quality indicator scale. The mapping from receiver metrics onto these levels is an implementation characterisation, not a specified formula.

Value	Interpretation	Value	Interpretation
0x0	Tone quality high	0x2	Tone quality low
0x1	Tone quality medium	0x3	Tone quality unavailable

9 From per-channel phase to distance

The ranging entity receives, for each measured channel k and each antenna path p , the two reported phase correction terms. It forms

$$H[k] = \text{PCT}_I[k] \text{PCT}_R[k] \implies \arg H[k] = \theta_I(f_k) + \theta_R(f_k) = \Theta(f_k), \quad (22)$$

that is, the two-way phase is obtained by *multiplying* the two complex terms rather than by adding two unwrapped angles. This is not a stylistic preference: multiplication in the complex plane is immune to the wrapping of the individual measurements, whereas adding two `atan2` outputs requires each to be unwrapped first, which is impossible for an isolated channel.

Figure 5 shows how the range estimators compose. As in Mode-1, a practical design is a chain: a robust ambiguity-free stage, a precise ambiguous stage, and a bias-correcting stage.

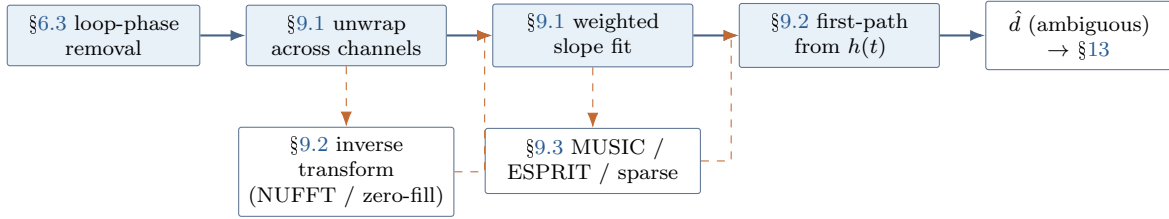


Figure 5: Range-estimator composition. Solid path: the common calibrate-unwrap-fit-correct chain. Dashed: transform-domain and super-resolution alternatives that replace or augment a stage.

9.1 Weighted phase-slope regression

Order the measured channels by frequency — they are visited in hopped order within the procedure [Core 6.3, Vol. 6, Part H, Sec. 4.2], so the first operation is a sort, not a fit — unwrap $\Theta(f_k)$ across the sorted list, and fit a straight line weighted by tone quality:

$$\frac{\widehat{\partial \Theta}}{\partial f} = \frac{\sum_k w_k (f_k - \bar{f}_w) (\Theta_k - \bar{\Theta}_w)}{\sum_k w_k (f_k - \bar{f}_w)^2}, \quad \hat{d} = -\frac{c}{4\pi} \frac{\widehat{\partial \Theta}}{\partial f}, \quad (23)$$

with w_k derived from the per-tone signal-to-noise estimate, and $\bar{f}_w$, $\bar{\Theta}_w$ the weighted means.

This is the exact frequency-domain dual of the Mode-1 group-delay regression: there the fit is over the 2 MHz of one packet, here it is over the full measured span. The estimators are identical; the apertures differ by a factor of about forty, and that factor is precisely the ratio of their resolutions and the reason one is unambiguous and the other is not.

Unwrapping is the fragile step. Sequential unwrapping fails whenever the true channel-to-channel phase step exceeds π , which by Eq. (16) happens beyond 74.95 m for adjacent 1 MHz channels but only 18.74 m across the raster’s built-in 4 MHz gap. It also fails on a single noisy or faded channel, and a single unwrap error propagates as a step to *all* subsequent channels, producing a large and confident-looking range error. Three practical countermeasures:

- Seed the unwrap with the Mode-1 range, so that the predicted phase at each channel is known to within a small fraction of a fringe and unwrapping becomes a nearest-integer decision rather than a sequential accumulation.
- Weight or exclude low-quality tones before unwrapping, not after.
- Prefer a wrapped-domain estimator that never unwraps at all: maximise

$$\left| \sum_k w_k H[k] e^{j4\pi f_k d/c} \right|$$

over a grid of d . This is the periodogram of the channel data in the delay domain, and it is exactly the inverse transform of Section 9.2 evaluated on an arbitrarily fine grid.

Residual as a diagnostic. The fit residual is as valuable as the slope. A residual that is flat and consistent with the tone quality estimates indicates a clean single-path channel; a systematically curved residual indicates uncorrected Φ_{loop} curvature; an oscillatory residual with a period in frequency of $1/\Delta\tau$ is the unmistakable signature of a second path at delay separation $\Delta\tau$ from the first.

9.2 Inverse transform and the channel impulse response

Because $H[k]$ is a set of frequency-domain samples of the channel (Eq. (19)), an inverse transform gives an estimate of the channel impulse response:

$$h(t) = \sum_k w_k H[k] e^{j2\pi f_k t}, \quad \hat{d} = \frac{c \hat{\tau}_{2way}}{2}, \quad (24)$$

where $\hat{\tau}_{2way}$ is the delay of the first significant peak of $|h(t)|$.

The properties of Eq. (24) follow directly from the sampling geometry:

Non-uniform channel sets. The CS channel list excludes some channels and the procedure may not visit every remaining one in every subevent [Core 6.3, Vol. 6, Part H, Sec. 4.2]. The resulting non-uniform sampling makes a plain inverse fast Fourier transform inapplicable. Three remedies, in increasing order of cost and fidelity: zero-fill the missing bins and accept the raised sidelobes; interpolate the missing bins from the fitted model and accept the resulting bias toward that model; or evaluate Eq. (24) directly on a delay grid as a non-uniform transform, which is exact and, for the few hundred delay bins of interest, entirely affordable.

Table 8: Transform-domain properties for the full 72-channel CS raster ($S = 74$ MHz, 1 MHz pitch, one 4 MHz gap).

Property	Expression	Value ($K = 72$, $S = 74$ MHz)
Delay resolution	$1/S$	13.5 ns
Range resolution	$c/(2S)$	2.03 m
Unambiguous delay span	$1/\Delta f$	$1.00 \mu s$
Unambiguous range span	$c/(2\Delta f)$	149.90 m
Safe sequential-unwrap range	$c/(4\Delta f_{max})$	18.74 m (4 MHz gap)
Sidelobe level, no window	-13.3 dB	—
Sidelobe level, Hann window	-31.5 dB, $1.5\times$ wider main lobe	—

Windowing is a bias–resolution trade. An unwindowed transform has -13.3 dB side-lobes, so a strong reflector at $+30$ ns can raise a sidelobe above a weak first path and capture the peak search. A Hann window suppresses this at the cost of a 50% wider main lobe, which *increases* the bias when two paths are genuinely unresolved. The correct choice depends on the channel and is worth making adaptively, on the observed dynamic range of $|h(t)|$.

First-path detection. Exactly as in Mode-1, the first peak matters and the largest peak does not. Back-search thresholding and constant-false-alarm-rate leading-edge detection transfer directly, with the threshold referenced to the sidelobe floor of the chosen window rather than to a correlation sidelobe level.

Slope fitting and transform peaks are not the same estimator. A weighted slope fit returns the amplitude-weighted mean delay of all the paths present; the transform peak search returns the delay of a selected path. In a clean single-path channel they agree, and the slope fit is more precise. In multipath they disagree, and the slope fit is *wrong* in a way that no amount of averaging repairs. A good implementation computes both and uses their disagreement as its multipath indicator.

9.3 Super-resolution methods

Treating $H[k]$ as a uniformly sampled sum of complex exponentials in the frequency variable makes the classical super-resolution machinery available: MUSIC and root-MUSIC on the covariance formed by spatial smoothing over frequency sub-apertures, ESPRIT on the shift-invariance of the same sub-apertures, and sparse recovery — orthogonal matching pursuit or an ℓ_1 formulation — over a dictionary of delay atoms.

These methods separate paths closer than $1/S$, which is where the 2.03 m resolution of Table 8 becomes the limiting factor in an indoor room. They come with three conditions that are frequently understated. They need an estimate of the model order, and they degrade sharply when it is wrong. They need either multiple snapshots or spatial smoothing, and smoothing spends aperture, so a 72-channel set smoothed into sub-apertures of 48 has an effective resolution set by about 49 MHz rather than 74 MHz. And they are far more sensitive than a slope fit to residual Φ_{loop} curvature, because a calibration error that a linear fit absorbs into its intercept appears to a subspace method as an additional spurious component.

Unlike Mode-1, where super-resolution over a single 2 MHz packet buys almost nothing, Mode-2 has the aperture to make these methods genuinely useful. They are the natural home of the accuracy that distinguishes a good phase-based implementation from an adequate one.

9.4 Comparison of range estimators

Table 9: Range-estimator comparison for the Mode-2 observable.

Method	Strength	Weakness	Typical role
Weighted slope fit (§9.1)	Attains the bound in a clean channel; cheap	Needs unwrapping; multipath-biased	Default estimator
Wrapped-domain periodogram	No unwrapping; ambiguity explicit	Grid search cost	Robust alternative
Zero-filled inverse FFT (§9.2)	Fast; shows the channel	Sidelobes from gaps	Diagnosis, first path
Non-uniform transform	Exact for any channel set	Cost scales with grid	Reference implementation
Windowed transform + CFAR	Robust first-path detection	Resolution loss from window	Multipath channels
MUSIC / root-MUSIC (§9.3)	Resolves below $1/S$	Model order; smoothing cost	High-accuracy mode
ESPRIT	Closed-form, no search	Sensitive to calibration error	High-accuracy mode
Sparse recovery	Handles gaps naturally	Regularisation tuning	Post-processing
Mode-1-seeded unwrap (§13)	Removes ambiguity entirely	Requires valid RTT	Always, where available

10 Precision: the Cramér–Rao analysis

10.1 Phase precision in one tone slot

For a single tone of energy $\mathcal{E}$ in white noise of one-sided density N_0 , the phase estimation variance is bounded by

$$\sigma_\theta \geq \frac{1}{\sqrt{2\mathcal{E}/N_0}}, \quad \frac{\mathcal{E}}{N_0} = \gamma B_n T_{PM}, \quad (25)$$

where γ is the signal-to-noise ratio measured in a reference noise bandwidth B_n . Taking $B_n = 1$ MHz and $T_{PM} = 10 \mu s$ gives a processing gain of exactly 10, so a 20 dB in-band signal-to-noise ratio yields $\mathcal{E}/N_0 = 30$ dB and $\sigma_\theta = 0.0224$ rad = 1.28° .

The two-way phase is the sum of two independent measurements, so

$$\sigma_\Theta = \sqrt{\sigma_{\theta,I}^2 + \sigma_{\theta,R}^2} = \sqrt{2} \sigma_\theta \quad (\text{symmetric links}). \quad (26)$$

10.2 Ranging precision from the slope fit

For K channels at frequencies f_k with equal weights, the slope variance of the fit Eq. (23) is $\sigma_\Theta^2 / \sum_k (f_k - \bar{f})^2$, so

$$\sigma_d = \frac{c}{4\pi} \frac{\sigma_\Theta}{\sqrt{\sum_k (f_k - \bar{f})^2}} = \frac{c}{4\pi} \frac{\sqrt{2} \sigma_\theta}{S} \sqrt{\frac{12(K-1)}{K(K+1)}}, \quad (27)$$

the second form holding for K channels spread uniformly over a span S . Three levers appear, and only three: the frequency span S , the number of channels K , and the per-tone signal-to-noise ratio through σ_θ . Span is the strongest, entering linearly; channel count and signal-to-noise ratio each enter as a square root.

Table 10: Cramér–Rao ranging standard deviation from Eqs. (25)–(27), with $B_n = 1$ MHz, evaluated on the actual CS channel raster. γ is the per-tone in-band signal-to-noise ratio; the unwrap limit is set by the largest gap in each subset, not by the pitch. These are bounds on the *random* error only; they say nothing about the calibration and multipath biases that dominate Table 12.

γ	T_{PM}	K	Span S	σ_θ	Unwrap limit	σ_d
30 dB	10 μs	72	74 MHz	0.41°	18.74 m	1.29 mm
20 dB	10 μs	72	74 MHz	1.28°	18.74 m	4.07 mm
10 dB	10 μs	72	74 MHz	4.05°	18.74 m	12.9 mm
0 dB	10 μs	72	74 MHz	12.8°	18.74 m	40.6 mm
20 dB	20 μs	72	74 MHz	0.91°	18.74 m	2.87 mm
20 dB	40 μs	72	74 MHz	0.64°	18.74 m	2.03 mm
20 dB	10 μs	36	73 MHz	1.28°	14.99 m	5.74 mm
20 dB	10 μs	18	71 MHz	1.28°	10.71 m	8.11 mm
10 dB	10 μs	18	71 MHz	4.05°	10.71 m	25.7 mm

Reading Table 10. Compare the 20 dB, 72-channel row — 4.3 mm — with the corresponding Mode-1 figure of 8.6 cm for 32 averaged steps at the same signal-to-noise ratio. Phase-based ranging is roughly twenty times more precise for the same air time, and that is the entire reason the mode exists. The comparison also shows where the effort must go: at a thermal floor of a few millimetres, a 0.1 ns loop-delay calibration residual (1.5 cm) and a 0.05 ppm Mode-0 residual (1.4 mm at $\Delta T = 190 \mu s$) are no longer negligible corrections but the leading terms. Note also the sixth and seventh rows: halving the channel count costs only $\sqrt{2}$ in precision but halves the unwrap limit, which is an availability failure rather than an accuracy one.

11 Reference estimation procedure for one Mode-2 step

The following is a complete implementation path, not a mandated algorithm.

- P1.** Compute the expected carrier for this step from the Mode-0 subevent reference and the channel’s nominal centre [Core 6.3, Vol. 6, Part A, Sec. 3.5]; programme the local oscillator once for the whole tone sequence and do not retune, restart or re-lock it between slots.
- P2.** Derive the slot schedule from T_{SW} , T_{PM} , N_{AP} and the step role; derive the antenna path permutation for this step and record it with the measurement.
- P3.** For each slot, open the measurement window strictly inside T_{PM} , with a guard at both ends that covers switch settling and the peer’s timing tolerance; capture $z[n]$ and record the gain state.
- P4.** Estimate the residual frequency offset jointly across the slots of the step, or apply the subevent value from Mode-0; do not re-estimate it independently per slot.
- P5.** Derotate and form the coherent average, Eq. (20); retain amplitude, phase, and the residual power after removing the fitted tone.

- P6.** Measure the tone extension slot with the same gain state and derive the noise and interference reference for this step.
- P7.** Compute the tone quality indicator from amplitude, residual and noise reference; mark a slot unavailable rather than reporting a low-confidence phase as if it were good.
- P8.** Scale and quantise the complex amplitude to the reported PCT format only at the interface boundary; retain full precision internally.
- P9.** Emit the record {step, channel, role, antenna path permutation, per-path PCT, per-path TQI, extension-slot indication, gain state, estimated δf , flags} [Core 6.3, Vol. 4, Part E, LE CS Subevent Result].
- P10.** At the ranging entity: pair the initiator and reflector terms per channel and path, form $H[k]$ by complex multiplication (Eq. (22)), apply $\hat{\Phi}_{loop}$ (Eq. (11)) and the $\epsilon\Delta T$ correction (Eq. (12)), sort by frequency, and run the estimator chain of Figure 5.

12 Aggregation across steps, paths and subevents

12.1 What is aggregated

Mode-2 differs from Mode-1 in the level at which combination happens. A Mode-1 step yields a complete range estimate that is then averaged. A Mode-2 step yields *one frequency-domain sample* per antenna path; the estimate exists only once the samples are assembled. Aggregation therefore has two distinct layers.

1. **Within an assembly.** Channels measured within one coherent interval are combined *coherently*, as the input to Eq. (23) or (24). This is where the aperture — and therefore the precision — comes from, and it requires that the loop phase, the frequency offset and the geometry be stable across the whole set.
2. **Across assemblies.** Successive subevents each yield a range estimate; these are combined *incoherently*, by weighted mean, median or a tracking filter.

The coherence interval is a physical limit, not a configuration choice. Coherent assembly is valid only while the geometry is stable to a small fraction of a wavelength. At 2.44 GHz the wavelength is 12.3 cm, so a tenth of a wavelength is 1.2 cm; a relative velocity of 1 m/s covers that in 12 ms. A procedure whose channel sweep takes longer than the coherence interval must either segment the sweep and combine the segments incoherently, or estimate and remove the range rate as part of the fit — which is a joint estimation of delay and Doppler, not a phase-slope fit.

12.2 Per-tone validity gating

A tone should be gated out or heavily down-weighted when: the reported tone quality indicator is low or unavailable in either direction; the tone amplitude is inconsistent with the extension-slot noise reference; the residual after removing the fitted tone is large; the per-slot estimated δf deviates from the subevent value; the gain state changed within T_{PM} ; clipping occurred; the antenna path permutation could not be resolved; or the peer did not report a term for that channel and path.

12.3 Weighted combination and robust alternatives

Within an assembly, weights enter the fit directly through w_k . Across assemblies, with $\hat{d}_m$ the estimate from assembly m and σ_m its predicted uncertainty,

$$w_m = g_m \min\left(w_{max}, \frac{1}{\sigma_m^2 + \sigma_{floor}^2}\right), \quad \hat{d} = \frac{\sum_m w_m \hat{d}_m}{\sum_m w_m}, \quad (28)$$

with the floor σ_{floor} representing the irreducible calibration and multipath contribution and the cap w_{max} preventing a single optimistic assembly from dominating.

Where the Mode-1 report recommends a lower-quantile combiner because multipath bias there is one-sided and positive, that recommendation does *not* transfer to Mode-2. A phase-slope estimate under multipath is an amplitude-weighted mean delay and can be biased in either direction; a transform-domain first-path estimate is biased long, like Mode-1, but only after it has correctly identified a first path. The right robust statistic is therefore the median rather than a low quantile, and the more valuable robustness comes from the multipath *indicator* of Section 9.2 — the disagreement between the slope fit and the transform peak — used to switch estimators rather than to trim samples.

13 Ambiguity resolution and the relationship to Mode-1 and Mode-3

13.1 The complementary pair

Table 11: Complementary character of the two ranging observables. The Mode-1 column reproduces the companion report’s figures for the same signal-to-noise ratio.

Property	Mode-1 (RTT)	Mode-2 (phase slope)
Observable	Packet arrival time	Channel phase $\Theta(f)$ versus f
Aperture	One packet, ≈ 2 MHz	Hopped band, up to 74 MHz
Resolution	≈ 500 ns, 75 m	≈ 13.5 ns, 2.03 m
Thermal precision at 20 dB	8.6 cm (32 steps)	4.1 mm (72 channels)
Ambiguity	None over the operating range	Periodic, 149.90 m
Dominant error	Loop delay bias; thermal jitter	Loop phase response; Mode-0 residual; multipath
Multipath character	One-sided, biased long	Two-sided for slope; long for first path
Needs from Mode-0	≈ 1 ppm	≈ 0.1 ppm
Integrity	Unpredictable waveform, NADM	None intrinsic; see §14

13.2 How the ambiguity is actually resolved

At the 1 MHz Bluetooth Low Energy channel raster the phase observable repeats every 149.90 m, which any usable Mode-1 estimate clears by two orders of magnitude. Ambiguity resolution in the strict sense — choosing the correct fringe — is therefore not the difficult part of Mode-2 in a standard configuration. Two related problems are the difficult part, and they are frequently conflated with it.

Unwrap seeding. Sequential unwrapping across channels fails on gaps and on faded tones long before the 149.90 m limit is approached, and each failure is a discrete range error. Seeding the unwrap with the Mode-1 range converts a fragile sequential accumulation into K independent nearest-integer decisions, each of which is safe provided the Mode-1 uncertainty is well below $c/(4f_{gap})$ for the largest gap present. This is the operationally important use of Mode-1.

Path selection. Choosing which peak of $|h(t)|$ is the direct path is a decision the phase observable cannot make alone in a strongly reflective channel. Mode-1's estimate, being an independent measurement with a different failure mode, bounds the search window.

The fusion is hierarchical, not symmetric. Mode-1 supplies a coarse, unambiguous, attack-resistant scale; Mode-2 supplies precision within it. The correct architecture computes the Mode-1 range first, uses it to seed unwrapping and to window the first-path search, and reports the Mode-2 result with the Mode-1 result retained as a consistency check. A Mode-2 range that disagrees with Mode-1 by more than the combined uncertainty is not a more precise answer; it is a detected fault — an unwrap slip, a permutation error, a wrong fringe, or an attack.

13.3 Mode-3

Mode-3 performs a CS_SYNC exchange and a tone exchange within a single step [Core 6.3, Vol. 6, Part H, Sec. 4.3.4]. Both interlude periods appear, and the step is correspondingly longer. The advantage is that the round-trip-time and phase observables for a given channel are acquired from the *same* step, so they share the same geometry, the same gain state and very nearly the same instant — which removes the motion and gain-state mismatch that can otherwise degrade the seeding described above. Everything in this report applies unchanged to the tone portion of a Mode-3 step; everything in the companion Mode-1 report applies unchanged to the packet portion.

14 Integrity considerations

The security architecture of Channel Sounding rests on the round-trip-time modes. A CS_SYNC packet carries a per-step access address derived from the CS deterministic random bit generator and, optionally, a random or marker-bearing sequence, so an attacker cannot synthesise a valid early arrival; the Normalized Attack Detector Metric quantifies departures from the expected waveform [Core 6.3, Vol. 6, Part H, Sec. 4.4].

A Mode-2 tone has none of these properties. It is an unmodulated carrier at a known frequency, fully predictable in advance, carrying no unpredictable content. An attacker can transmit a tone of arbitrary phase and amplitude; the receiver has no intrinsic means to distinguish it from the legitimate one. Phase-based ranging is consequently *not* a distance-bounding measurement, and a Mode-2 result should never be treated as one on its own.

Three practical consequences:

- A security-relevant ranging decision must be gated on the Mode-1 integrity metrics of the same procedure, not only on Mode-2 precision.
- The channel hopping sequence is itself a defence, since an attacker must track it to inject coherently on every channel; the randomisation of that sequence and of the antenna path permutation should not be weakened for implementation convenience.
- The Mode-1 versus Mode-2 consistency check of Section 13 is a detector as well as a diagnostic, and its threshold should be set with that role in mind.

15 Worked numerical example at 2 440 MHz

Assume antenna configuration index 0 ($N_{AP} = 1$), $T_{SW} = 10 \mu s$, $T_{PM} = 10 \mu s$, $T_{RD} = 5 \mu s$, $T_{IP2} = 145 \mu s$, a coherent assembly of all $K = 72$ CS channels — $2402+k$ MHz for $k = 2 \dots 22$ and $26 \dots 76$, spanning $S = 74$ MHz with one 4 MHz gap — a per-tone in-band signal-to-noise ratio of $\gamma = 20$ dB, and a true separation of $d = 10.000$ m.

15.1 Nominal timing

From Eqs. (1) and (2),

$$(T_{SW} + T_{PM})(N_{AP} + 1) = 20 \times 2 = 40 \mu s, \quad (29)$$

$$T_{step,2} = 2(40 + 5) + 145 = 235 \mu s, \quad (30)$$

$$\Delta T = 40 + 5 + 145 = 190 \mu s. \quad (31)$$

15.2 The quantity being measured

$$\tau = \frac{10.000}{299\,792\,458} = 33.356410 \text{ ns}, \quad (32)$$

$$\Theta(f_c) = -\frac{4\pi f_c d}{c} = -2\pi \times 162.77928 \text{ rad} \equiv 4.896351 \text{ rad} = 280.538^\circ \pmod{2\pi}. \quad (33)$$

The measured phase on any single channel is therefore uninformative on its own: the integer part 162 is unknown. What carries the distance is the *change* of Θ across channels,

$$\Delta\Theta = \frac{4\pi d \Delta f}{c} = 0.4191690 \text{ rad} = 24.0166^\circ \text{ per 1 MHz}, \quad (34)$$

and the sensitivity is 0.41637 m of range per degree of error in that per-channel step.

15.3 Aperture properties

$$\text{range resolution } c/(2S) = 2.0256 \text{ m}, \quad (35)$$

$$\text{ambiguity period } c/(2\Delta f) = 149.896 \text{ m}, \quad (36)$$

$$\text{adjacent-channel unwrap limit } c/(4\Delta f) = 74.948 \text{ m}, \quad (37)$$

$$\text{unwrap limit at the raster's 4 MHz gap} = 18.737 \text{ m}. \quad (38)$$

15.4 Thermal limit

From Eq. (25) with $\mathcal{E}/N_0 = 100 \times 10^6 \times 10^{-5} = 1000$:

$$\sigma_\theta = \frac{1}{\sqrt{2000}} = 0.0223607 \text{ rad} = 1.2812^\circ, \quad (39)$$

$$\sigma_\Theta = \sqrt{2} \sigma_\theta = 0.0316228 \text{ rad} = 1.8117^\circ, \quad (40)$$

$$\sum_k (f_k - \bar{f})^2 = 3.4445 \times 10^{16} \text{ Hz}^2 \quad (\text{actual raster}), \quad (41)$$

$$\sigma_d = \frac{c}{4\pi} \frac{\sigma_\Theta}{\sqrt{3.4445 \times 10^{16}}} = 4.065 \text{ mm}. \quad (42)$$

15.5 Frequency-offset term

From Eq. (13) with $\Delta T = 190 \mu s$: an uncorrected 12 ppm oscillator pair gives

$$\delta d_\epsilon = \frac{c \times 12 \times 10^{-6} \times 190 \times 10^{-6}}{2} = 34.18 \text{ cm}, \quad (43)$$

while a Mode-0 correction good to 0.05 ppm reduces it to 1.42 mm. Note that this is a bias common to every channel of the assembly, so it is not reduced by adding channels; only the Mode-0 correction reduces it.

15.6 Loop phase response

An uncalibrated loop group delay of 10 ns biases the range by

$$\delta d = \frac{c \times 10 \text{ ns}}{2} = 1.499 \text{ m}, \quad (44)$$

which exceeds every other term by two orders of magnitude. Calibrated to a residual group delay of 0.1 ns the contribution falls to 1.50 cm — still the largest single entry in the budget below, and roughly three times the thermal floor.

15.7 Reporting quantum

With the phase correction term reported as a 12-bit signed in-phase and quadrature pair, a full-scale term has a phase quantisation standard deviation of about $(1/\sqrt{12})/2047 = 1.41 \times 10^{-4}$ rad. Propagated through Eq. (27) this contributes 0.027 mm, four orders of magnitude below the loop-delay term: the reporting format is not a limiting factor, provided the internal scaling actually uses the available range rather than a small fraction of it.

15.8 Assembled budget

Table 12: Assembled uncertainty for the worked example: $N_{AP} = 1$, $T_{PM} = 10 \mu s$, $\gamma = 20$ dB, all $K = 72$ CS channels spanning 74 MHz, $T_{IP2} = 145 \mu s$. The root-sum-square is an upper-bound convenience only: the rows marked as biases are common to the whole assembly and are neither reduced by adding channels nor correctly combined in quadrature.

Contribution	Phase	Range	Character
Loop group-delay calibration residual (0.1 ns)	—	1.499 cm	bias, per channel/path
Thermal noise, 72 channels	1.81° per tone	0.407 cm	random
Oscillator phase noise (0.5° rms)	0.71° per tone	0.159 cm	partly correlated
Mode-0 residual, 0.05 ppm at $\Delta T = 190 \mu s$	—	0.142 cm	bias, per assembly
FAE table quantisation, 0.0156 ppm	—	0.044 cm	bias, per channel
PCT quantisation, 12-bit I/Q	0.008°	0.003 cm	random
Root-sum-square (excluding multipath)	—	1.568 cm	
Multipath, indoor, slope estimator	—	± 10 to 50 cm	two-sided, environment
Multipath, indoor, first-path estimator	—	typ. +5 to 20 cm	one-sided, environment

The table carries the report’s central practical message, and it is the same message as the Mode-1 report reaches from the other direction. Under favourable conditions the instrumental terms combine to about a centimetre and a half, dominated not by the estimator but by the loop calibration residual. The residual multipath error, which is not reduced by adding channels, is then an order of magnitude larger than everything else. Effort spent on

loop-phase calibration and on transform-domain first-path detection buys far more accuracy than effort spent on the per-slot phase estimator.

Table 13: Deterministic values for the 2 440 MHz example.

Quantity	Value	Unit
Antenna configuration index	0 ($N_{AP} = 1$)	
$T_{SW} / T_{PM} / T_{RD} / T_{IP2}$	10 / 10 / 5 / 145	μs
Tone sequence duration	40	μs
Step duration $T_{step,2}$	235	μs
Reciprocal measurement interval ΔT	190	μs
Channels K / pitch / span	72 / 1 / 74	—, MHz, MHz
Largest gap in the CS raster	4	MHz
True distance	10.000	m
One-way propagation delay	33.356410	ns
Two-way phase at f_c , modulo 2π	4.896351	rad
Two-way phase step per 1 MHz	0.4191690	rad
Two-way phase step per 1 MHz	24.0166	deg
Range per degree of per-channel phase step	0.41637	m
Range resolution $c/2S$	2.0256	m
Ambiguity period $c/2\Delta f$	149.896	m
Adjacent-channel unwrap limit $c/4\Delta f$	74.948	m
Unwrap limit at the 4 MHz raster gap	18.737	m
σ_θ at $\gamma = 20$ dB, $T_{PM} = 10 \mu s$	0.0223607	rad
σ_Θ (two-way)	0.0316228	rad
CRLB σ_d , 72 channels	4.065	mm
Bias from 10 ns uncalibrated loop delay	1.499	m
Bias from 12 ppm uncorrected offset at $\Delta T = 190 \mu s$	0.342	m
Bias from 0.05 ppm residual at $\Delta T = 190 \mu s$	1.424	mm

16 Error sources and uncertainty

For approximately independent contributions the first-order budget on the two-way phase is

$$\sigma_\Theta^2 \approx \sigma_{th}^2 + \sigma_{pn}^2 + \sigma_{dc}^2 + \sigma_{iq}^2 + \sigma_q^2, \quad (45)$$

propagated to range through Eq. (27), with the loop-phase residual, the frequency-offset term and the multipath term handled separately because they are biases rather than variances. Correlated terms must not be combined by blind root-sum-square: the loop-phase calibration residual is common to every channel of an assembly and the frequency-offset term is common to the whole subevent, so both survive the $\sqrt{K}$ reduction that applies to σ_{th} .

Table 14: Representative Mode-2 phase and ranging error budget.

Source	Effect	Mitigation	Diagnostic
Thermal noise	Random phase error $1/\sqrt{2\mathcal{E}/N_0}$ per tone	Longer T_{PM} , more channels, wider span	Scatter versus predicted CRLB
Loop group delay	Range bias $cD_{loop}/2$	Conducted calibration per channel, path, gain state	Cable loopback at known delay
Loop phase curvature	Slope-fit residual; smeared transform peak	Full phase-response calibration surface, not a scalar	Residual shape versus frequency
Residual frequency offset	Bias $c \epsilon \Delta T/2$, linear in f	Mode-0; short T_{IP2} and tone sequence	Range versus T_{IP2} sweep
In-slot frequency error	Phase bias $\pi \delta f T_{PM}$	Joint δf estimate; derotation	Phase versus injected offset
Oscillator phase noise	Random phase, correlated within a step	Integrated phase-noise budget over T_{PM}	Allan-type variance of Θ
Receiver DC offset	Vector added to the tone estimate	Non-zero IF placement; extension-slot subtraction	Phase versus tone amplitude
I/Q imbalance	Image-induced phase error, δf -dependent	Per-gain-state image calibration	Phase versus δf sweep
AGC state change	Path change within T_{PM}	Freeze gain across the tone sequence	Phase versus input level
LO retune between slots	Loss of common phase reference across paths	Continuous LO over the whole sequence	Inter-path phase repeatability
Switch settling inside T_{PM}	Slot-dependent phase bias	Adequate T_{SW} ; measure settling	Phase versus T_{SW} setting
Antenna permutation error	Paths attributed to wrong geometry	Verify against asymmetric layout	Per-path range consistency
Channel gaps	Unwrap slips; transform sidelobes	Mode-1-seeded unwrap; non-uniform transform	Residual versus gap size
Unwrap slip	Discrete range jump	Wrapped-domain estimator; seeding	Mode-1/Mode-2 disagreement
Multipath, slope estimator	Two-sided bias toward weighted mean delay	Transform-domain first-path detection	Slope versus peak disagreement
Multipath, first-path estimator	One-sided positive bias	CFAR leading edge; super-resolution	Peak-to-first-path offset
Motion within an assembly	Phase ramp in time misread as range	Segment the sweep; joint delay-Doppler fit	Range rate consistency
Interference on one channel	Corrupted tone; biased fit	Extension-slot noise reference; gating	Per-channel rejection rate
Tone injection attack	Arbitrary forged phase	Gate on Mode-1 integrity; hopping	Mode-1/Mode-2 disagreement

17 Verification and conformance-oriented test plan

17.1 Layered verification

1. **Pure DSP vectors.** Synthesised tone slots at exactly known phases, swept over a full 2π , with additive white Gaussian noise, carrier offset, DC offset, I/Q imbalance and clipping. Verify that the estimated phase tracks the injected phase with bounded residual, and characterise the estimator's bias as a function of δf and of DC offset.
2. **Synthetic channel vectors.** Construct $H[k]$ analytically for a known single-path delay across the full channel set, including realistic gaps, and verify that the slope fit and the transform peak both return the injected delay. Then repeat with an injected $\Phi_{loop}(f)$ of known shape and verify that the calibration removes it.
3. **Multipath vectors.** Two- and three-path channels with controlled amplitude ratios and separations from $0.1/S$ to $3/S$. Verify the sign and magnitude of the slope-fit bias — it must be shown to go both ways — and the first-path estimator's bias and failure threshold.
4. **Unwrap stress vectors.** Sweep true distance from 0 to beyond $c/(4\Delta f)$ and verify the unwrap failure point matches Eq. (16); repeat with the largest permitted channel gap and confirm the reduced limit. Verify that Mode-1 seeding restores correct operation.
5. **Conducted delay-line tests.** Two devices connected through calibrated cable lengths or a programmable delay line. This is the only test that measures Φ_{loop} directly; it must be swept over channel, antenna path, gain state and temperature, and the result stored as a phase response rather than a scalar.
6. **Path-reversal test.** Repeat the conducted test with initiator and reflector roles exchanged; a difference exposes non-reciprocal hardware phase.
7. **Frequency-offset tests.** Deliberately offset one device's reference by a known ppm value and verify that the measured range shifts by $c \epsilon \Delta T/2$ and that applying the Mode-0 result removes it. Sweep the sign to catch an inverted correction, and sweep T_{IP2} across its permitted set: a residual slope of range versus T_{IP2} is the unambiguous signature of an uncorrected or wrongly signed frequency term.
8. **Antenna path tests.** With $N_{AP} > 1$ and a deliberately asymmetric antenna layout, verify that each path's range matches its geometry and that the permutation is applied correctly. Verify that the local oscillator remains phase-continuous across the sequence by checking inter-path phase repeatability across steps.
9. **Tone extension slot tests.** Verify the noise reference with a known injected interferer, and verify correct handling of each configured extension-slot option.
10. **Anechoic and multipath chamber tests.** Absolute accuracy at known geometric distances, then in a reverberant environment, to separate instrumental error from channel bias.
11. **Motion tests.** Constant-velocity runs at increasing speed to find the coherence limit of the assembly, and verification that the segmentation or delay-Doppler strategy holds accuracy beyond it.
12. **Integrity tests.** Tone injection at controlled phase offsets; verify that the Mode-1/Mode-2 consistency check detects the induced disagreement at the intended threshold.

17.2 Minimum acceptance records

Retain: RF configuration, CS channel list and the channels actually measured, antenna configuration index and per-step permutation, T_{SW} , T_{PM} , T_{IP2} and T_{FCS} settings, the applied FFO_{RX}^* , the Φ_{loop} calibration surface version, per-channel and per-path raw PCT and TQI, extension-slot indications and noise references, the assembled $H[k]$, both the slope-fit and transform-domain range estimates with their residuals, the unwrap decisions and their seeds, the Mode-1 range used for seeding, aggregated result with accepted channel count and scatter, and pass/fail limits. Reproducible vectors should include the worked example of Section 15, a full- 2π phase sweep, a known two-path case with the reflection both stronger and weaker than the direct path, an injected frequency offset of each sign, and a channel set with a deliberate gap.

18 Normative behaviour versus implementation choice

Table 15 separates what the specification fixes from what the implementer decides. The pattern is the same as in Mode-0 and Mode-1, but the balance is more extreme: the specification fixes the air interface and the reported observable, and leaves the whole path from observable to metres — which in Mode-2 is most of the accuracy — to the implementer [Core 6.3, Vol. 1, Part A, Sec. 9.2].

Table 15: Boundary between specified behaviour and implementation design.

Bluetooth-defined behaviour	Implementation-specific design
Mode-2 step structure: tone slots, ordering, $N_{AP} + 1$ slots [Core 6.3, Vol. 6, Part H, Sec. 4.3.3]	Receive window placement and guards within T_{PM}
T_{SW} , T_{PM} , T_{RD} , T_{IP2} , T_{FCS} and their permitted sets	Which permitted values are chosen, and the resulting ΔT trade
Tone is unmodulated at the channel centre frequency [Core 6.3, Vol. 6, Part A, Sec. 3.5]	Intermediate-frequency placement, sample rate, numeric format
Antenna configuration index and path permutation [Core 6.3, Vol. 6, Part H, Sec. 4.5]	Per-path calibration, diversity, averaging or angle processing
Tone extension slot presence and indication	Use of the slot as a noise and interference reference
PCT reporting format and TQI scale [Core 6.3, Vol. 4, Part E]	Phase estimator; TQI computation and thresholds
Requirement that measurements be referenced consistently between devices	Choice of antenna-plane reference and its calibration
Use of the Mode-0 result for frequency compensation	Rescaling arithmetic, precision, filtering of FFO_{RX}^*
Channel selection and hopping within the procedure [Core 6.3, Vol. 6, Part H, Sec. 4.2]	Assembly of channels into a coherent aperture; gap handling
Architectural role of phase and amplitude information [Core 6.3, Vol. 1, Part A, Sec. 9.2]	The entire ranging algorithm: unwrap, transform, first path, ambiguity, aggregation

Four conclusions that prevent common design errors. (1) The two-way construction cancels the unknown relative oscillator phase but *accumulates* the four hardware phase responses into a loop response whose slope is an additive range bias and whose curvature corrupts the fit; it must be calibrated as a function of frequency, not as a scalar. (2) A residual fractional frequency offset produces a phase term linear in frequency and therefore reads directly as distance, $c \epsilon \Delta T/2$; Mode-2 needs roughly ten times the Mode-0 accuracy that Mode-1 needs. (3) Multipath bias on a phase-slope estimate is two-sided, so the Mode-1 habit of trimming to a low quantile is actively wrong here; use the slope-versus-transform disagreement to switch estimators instead. (4) The local oscillator must be phase-continuous across the whole tone sequence, or the antenna paths do not share a reference and per-path phases are not comparable.

19 Conclusions

Mode-2 converts a reciprocal tone exchange into a precise distance. The measurement is structurally elegant for the same reason Mode-1 is: because each device measures phase against its own oscillator and both results are reported, the unknown relative oscillator phase cancels exactly and only the propagation phase survives. That elegance is bounded by three physical facts. The four hardware phase responses add rather than cancel, so the accuracy of a Mode-2 range is ultimately the accuracy of a phase calibration over frequency. The two measurements are separated in time, so any residual frequency error is integrated over that separation and appears as range. And phase is periodic, so the estimate is precise but not, by itself, unique.

The estimation problem splits into a per-slot problem and a cross-channel problem, and the second is where implementations differ. Per-slot phase estimation is a well-posed single-tone problem at comfortable signal-to-noise ratio; the cross-channel problem is spectral estimation on wrapped, hopped, non-uniformly sampled data, where the difference between a naive sequential unwrap and a seeded wrapped-domain estimator is the difference between occasional large errors and none. The Cramér–Rao analysis ties the achievable precision to just three quantities — frequency span, channel count and per-tone signal-to-noise ratio — and shows that the thermal floor at practical settings is a few millimetres, an order of magnitude below the calibration and multipath terms that then dominate.

Mode-2 is not a complete ranging solution on its own, and it is not intended to be. It supplies precision within the coarse, unambiguous, attack-resistant scale that the round-trip-time modes provide. A scientifically complete implementation validates the phase chain and the timing chain separately, and then validates their combination — including the case where they disagree, which is the most informative measurement the system can make.

A Reported quantities at the host interface

Per Mode-2 step, the controller reports a small set of fields to the host in the CS subevent result. Their exact names, widths and encodings are defined in Core 6.3, Volume 4, Part E; the summary below is given to make the data flow concrete and should be checked against the specification revision in force.

Three implementation notes follow from the encoding. First, the reported term is a complex number in the measuring device’s own reference; it is not a phase in radians and it is not referenced to the peer. Forming a two-way phase requires both devices’ terms for the same channel and path, which is why the ranging entity must sit above both — a controller cannot compute a range from its own reports alone. Second, the 12-bit fields are signed and the internal scaling must exercise their range: a term scaled to a small fraction of full

Table 16: Per-step Mode-2 reported fields.

Field	Encoding	Role
Antenna permutation index	index	Maps tone slot to physical antenna path
PCT, per antenna path	12-bit signed I and Q	The measured complex tone amplitude
Tone quality indicator, per path	ordinal, Table 7	Gating and weighting input
Tone extension indication	flag / count	Whether the extension slot carried a tone
Packet / tone antenna	index	Multi-antenna calibration selection
Unavailable indicator	reserved value	Tone must be gated out

scale throws away signal-to-noise ratio that the receiver actually had. Third, the reserved “unavailable” value must be tested for explicitly; interpreting it as a numeric complex value injects a nonsense point into a coherent fit, and a single such point in a least-squares slope fit is enough to move the range by metres.

B Numerical test vectors

- B1. Phase sweep.** Synthesise a $T_{PM} = 10 \mu s$ tone slot with true phase swept over $[0, 2\pi)$ in 5° steps, noise-free, with $\delta f = 0$. The estimated phase must reproduce the injected phase to within the documented estimator bias.
- B2. Frequency-offset sensitivity.** Repeat B1 with $\delta f \in \{0, 1, 5, 10\}$ kHz and no derotation. The phase bias must follow $\pi \delta f T_{PM}$: 0, 0.0157, 0.0785 and 0.157 rad respectively. With derotation applied, the residual must fall below the documented limit.
- B3. Thermal consistency.** Repeat B1 at $\gamma = 0, 10, 20$ and 30 dB with 1 000 trials each. The sample standard deviation of θ must lie within a documented margin of 12.81° , 4.05° , 1.28° and 0.41° .
- B4. Single-path channel assembly.** Construct $H[k]$ for $k = 0 \dots 71$ at 1 MHz spacing with $d = 10.000$ m. The per-channel two-way phase step must be 24.0166° , the slope fit must return 10.000 m, and the transform peak must fall at 66.713 ns of two-way delay.
- B5. End-to-end thermal.** Add noise to B4 at $\gamma = 20$ dB over 1 000 trials. The standard deviation of the estimated range must be near 4.28 mm.
- B6. Loop phase.** Insert a known linear Φ_{loop} corresponding to $D_{loop} = 10.0$ ns and verify the reported range shifts by 1.499 m, and that the calibration removes it to the documented residual. Repeat with a quadratic component and verify that the slope-fit residual exposes it.
- B7. Frequency offset.** With $\Delta T = 190 \mu s$, inject $\epsilon = +12$ ppm. Uncorrected, the range must shift by +34.18 cm; with the Mode-0 result applied, the residual must fall below 2 mm. Repeat with $\epsilon = -12$ ppm to catch a sign inversion, and repeat at $T_{IP2} = 10 \mu s$ to confirm the ΔT scaling.
- B8. Unwrap limit.** Sweep d from 70 to 80 m and confirm that sequential unwrapping fails above 74.95 m for a hypothetical gapless 1 MHz set. Then repeat on the real CS

raster and confirm that failure begins instead at 18.74 m, set by its 4 MHz gap, while the Mode-1-seeded estimator does not fail.

- B9. Two-path bias.** Add a second component at +30 ns with -3 , 0 and $+3$ dB relative amplitude. Record the slope-fit bias — it must change sign across the set — and the transform first-path bias, which must remain positive.
- B10. Permutation.** With $N_{AP} = 4$ and four deliberately different geometric path lengths, verify that each path's estimated range matches its geometry under every permutation the implementation can emit.

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